

정답

About $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$

When $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \alpha$.

$$\lim_{x \rightarrow a} g(x) = 0 \Rightarrow \lim_{x \rightarrow a} f(x) = 0$$

$$\text{Q. } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{0}{0} \left(= \frac{\lim_{x \rightarrow a} f(x)}{0} \right) = \alpha.$$

type이 이상하?

$$\Rightarrow \lim_{x \rightarrow a} f(x) = 0.$$

아래 해준 음 type이 되겠!

$$\lim_{x \rightarrow a} f(x) = 0 \text{ AND } \alpha \neq 0 \Rightarrow \lim_{x \rightarrow a} g(x) = 0.$$

$$\text{Q. } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{0}{0} \left(= \frac{0}{\lim_{x \rightarrow a} g(x)} \right) = \alpha.$$

type이 이상하?

$$\Rightarrow \lim_{x \rightarrow a} g(x) = 0.$$